SRM BLOOD CONNECT WEBSITE

(MINOR PROJECT- UCS23P04L)

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by

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BONAFIDE CERTIFICATE

Certified that this project report titled **" SRM BLOOD CONNECT WEBSITE "** is the Bonafide work of MATHAVAN **V (REG NO: RA2331005050049)** who carried out the Major project work under my supervision at **SRM INSTITUTE OF SCIENCE AND TECHNOLOGY**, Tiruchirappalli Campus, in 2026.

INTERNAL GUIDE ***HEAD OF THE DEPARTMENT***

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

This project, “SRM Blood Connect,” addresses the critical problem of inefficient blood donor management and delayed emergency response in medical situations by providing a centralized web-based platform for connecting donors, recipients, and administrators in real time. The primary objective is to streamline blood donation processes, ensure quick access to compatible donors, and maintain an organized database for monitoring donor availability, requests, and donations. The system employs a modern web-based methodology using frontend interfaces and backend integration (such as real-time database services) to manage donor registration, alert generation, request tracking, and administrative control through a dedicated dashboard. The algorithm focuses on matching blood group compatibility, availability status, and urgency level to optimize donor-recipient connections. Findings indicate that such a system significantly reduces response time and improves coordination compared to traditional manual methods. The motivation behind this project stems from the frequent shortage of blood and lack of immediate access during emergencies, especially in local communities and institutions. Existing systems are often fragmented, lack real-time updates, or depend heavily on manual communication, whereas this project integrates automation, live tracking, and centralized control for better efficiency. The accuracy of the system depends on real-time data updates and proper donor input, ensuring reliable matching and alert delivery. Overall, this project provides a scalable, efficient, and user-friendly solution to enhance blood donation management and potentially save lives through faster and more accurate connections.

1. INTRODUCTION

Blood is one of the most vital components of the human body, playing an essential role in sustaining life by transporting oxygen and nutrients to various organs. During medical emergencies such as accidents, surgeries, childbirth complications, and critical illnesses, timely access to the correct blood group is crucial for survival. However, identifying and reaching suitable blood donors at the right time remains a significant challenge due to the absence of a centralized system, lack of real-time information, and dependence on manual communication methods such as phone calls and social media. These traditional approaches are often unreliable, time-consuming, and inefficient, leading to delays that can put patients' lives at risk.

In many situations, patients and their families struggle to find donors quickly, especially during urgent conditions. Existing blood donation systems are often fragmented, where donor information is either outdated or not easily accessible. Additionally, there is limited coordination among donors, which reduces the effectiveness of available resources. These issues highlight the growing need for a modern, digital solution that can efficiently connect donors and recipients while ensuring fast and reliable communication.

To address these challenges, the project "SRM Blood Connect" has been developed as a web-based platform aimed at simplifying and improving the blood donation process within the SRM community. This system provides a centralized digital platform where users such as students, faculty, and staff can register as blood donors, update their availability, and respond to emergency blood requests. By creating a connected network of donors, the platform ensures that help can be provided quickly during critical situations.

The system maintains a structured and unified database of verified donors, including their blood groups, contact details, and availability status. This organized approach enables users to easily search and identify suitable donors based on compatibility and urgency. By eliminating the need for manual searching and reducing dependency on informal communication methods, the platform significantly improves efficiency and reduces response time during emergencies.

SRM Blood Connect consists of two main components that work together to ensure smooth operation. The first component is the User Interface, which allows donors and recipients to register, access information, search for donors, and raise blood requests through a simple and user-friendly interface. The second component is the Admin Panel, which acts as the central control system. Administrators are responsible for monitoring activities, verifying donor

details, managing requests, and maintaining accurate records through a secure dashboard. This centralized management ensures reliability, transparency, and efficient functioning of the system.

From a technical perspective, the platform is developed using modern web technologies such as HTML, CSS, and JavaScript for the frontend, ensuring a responsive and visually appealing design. Supabase is used for backend services, including database management, authentication, and real-time data synchronization. This combination of technologies enables the system to provide instant updates, secure user access, and scalable performance.

One of the key advantages of SRM Blood Connect is its ability to provide real-time communication and updates. When a blood request is generated, the system helps in identifying and notifying suitable donors quickly, thereby minimizing delays. Additionally, the platform maintains proper records of donors, requests, and donations, which enhances transparency and supports better decision-making and future improvements.

The motivation behind this project arises from the frequent shortage of blood and the inefficiencies present in existing systems, especially within educational institutions and local communities. By introducing a simple and accessible digital platform, the project encourages voluntary blood donation and improves awareness among users. It also helps in building a reliable and responsive donor network over time.

In conclusion, SRM Blood Connect is a comprehensive and technology-driven solution designed to modernize the blood donation process. By integrating real-time data management, user-friendly interfaces, and centralized administration, the platform improves communication, reduces delays, and ensures better utilization of available resources. Ultimately, it aims to save lives by ensuring that the right blood reaches the right person at the right time in an efficient and reliable manner.

2. SYSTEM REQUIREMENTS

2.1 Hardware Requirements

Minimum Hardware Requirements

- **Processor:** Intel Core i3 / AMD equivalent
- **RAM:** 4 GB
- **Hard Disk:** 10 GB free space
- **Monitor:** 1024 × 768 resolution
- **Input Devices:** Keyboard and Mouse
- **Network:** Basic Internet connection

Recommended Hardware Requirements

- **Processor:** Intel Core i5/i7
- **RAM:** 8 GB or higher
- **Storage:** SSD with 20 GB free space
- **Monitor:** Full HD (1920 × 1080)
- **Network:** High-speed Internet

These specifications ensure smooth development, testing, and operation of the system.

2.2 Software Requirements

Operating System

Any one of the following:

- Windows 10 / 11
- Linux (Ubuntu / Fedora)
- macOS

Frontend Technologies

- **HTML5** – Page structure

- **CSS3** – Styling, layout, UI design
- **JavaScript** – Client-side logic and interactions

Backend / Database

- **Supabase** (PostgreSQL Database + Authentication)
    - For storing donor data
    - Real-time updates
    - Secure login for admin

Development Tools

- **Visual Studio Code (VS Code)** – Code editor
- **Live Server Extension** – For previewing the project
- **Browser:** Google Chrome (recommended), Firefox, Edge

Optional Tools

- Git & GitHub (Version control)
- Node.js (For optional JavaScript modules)

2.3 System Descriptions

The *SRM Blood Connect* system is divided into three main components:

1. User Interface (Frontend)

- Designed using HTML, CSS, and JavaScript
- Contains pages like:
    - Home page
    - Donor registration form
    - Blood request form
    - Login page
- Fully responsive and user-friendly

- Includes modern UI features such as animations, gradients, and custom cursor

2. Admin Panel

- Accessible only to authorized admin users
- Allows admin to:
    - View donor list
    - Approve or verify details
    - Manage blood requests
    - Monitor real-time updates
- Built with advanced UI components using CSS and JavaScript

3. Database & Backend

- Powered by Supabase
- Stores:
    - Donor information
    - Blood group details
    - Contact number
    - Request information
    - Login credentials
- Provides:
    - Real-time synchronization
    - Secure authentication
    - Easy data retrieval
    - Automatic timestamps

These components together ensure efficient functioning, secure access, and smooth data management for the entire system.

3. SYSTEM ANALYSIS

3.1 Feasibility Study

A feasibility study is conducted to evaluate whether the project is practical, workable, and beneficial. It includes the following analyses:

1. Technical Feasibility

- The project uses simple and widely supported technologies such as **HTML, CSS, JavaScript**, and **Supabase**.
- No advanced hardware or complex server setup is required.
- The system can be deployed on any device with a web browser.

Conclusion: The project is technically feasible.

2. Economic Feasibility

- Uses free and open-source tools.
- No licensing cost for Supabase (free tier).
- No server maintenance cost for testing and development.

Conclusion: The project is economically feasible.

3. Operational Feasibility

- Easy-to-use interface for students, staff, and admin.
- Reduces manual work and speeds up emergency communication.
- Can be operated with minimal training.

Conclusion: The project is operationally feasible.

4. Time Feasibility

- Development time is low due to lightweight web technologies.
- Project implementation can be completed within a short duration.

Conclusion: Time feasibility is strong.

3.2 Existing System

In the existing blood donation process within the campus:

- Students or staff manually search for donors.
- Donor details are shared through WhatsApp groups or phone calls.
- There is no verified or centralized donor database.
- Emergency requests often take longer to reach the right person.

The existing system is **manual, inconsistent, and time-consuming**.

3.3 Drawbacks of Existing System

1. **No Centralized Database:** Donor information is scattered and unorganized.
2. **Delayed Response:** Emergency blood requests may not reach the right donors on time.
3. **Unverified Data:** Donor availability and contact information may be outdated.
4. **Lack of Tracking:** No proper record of past donations or requests.
5. **High Dependency on Social Media:** WhatsApp and phone calls are unreliable during urgent situations.
6. **No Admin Monitoring:** No proper authority to approve or validate donor details.

3.4 Proposed System

The proposed *SRM Blood Connect* system introduces a digital platform with:

- **Online donor registration** with verified details.
- **Centralized database** storing all donor and request information.
- **Admin panel** for managing donors and requests.
- **Real-time updates** using Supabase backend.
- **User-friendly interface** with responsive design.

- **Secure login authentication** for admin users.

The system ensures that blood donors and recipients can connect quickly and reliably.

3.5 Benefits of Proposed System

1. **Faster Communication:** Donor details can be accessed instantly.
2. **Accurate & Centralized Data:** All records stored in one place.
3. **Verified Donor Information:** Admin controls validation and updates.
4. **Reduced Manual Work:** Automated data handling saves time.
5. **Improved Emergency Response Time:** Quick identification of suitable donors.
6. **User-Friendly Design:** Modern interface makes it easy for anyone to use.
7. **Real-Time Updates:** Immediate synchronization of donor and request entries.
8. **Scalability:** New modules or features can be added easily in future.

3.6 Scope of the Project

The scope of SRM Blood Connect includes:

- Building a digital blood donation platform for SRM campuses.
- Managing donors, blood groups, and emergency requests efficiently.
- Providing admin monitoring and real-time data updates.
- Enhancing communication between donors and recipients.

Future Expansion Scope

- Integration with hospitals and medical centers.
- Mobile app version (Android/iOS).
- SMS/Email alert system for urgent requests.
- Automatic donor-matching using AI.

4. SYSTEM DESIGN

4.1 Design and Architecture

The architecture of the SRM Blood Connect system is designed in a modular manner to ensure easy development, maintenance, and scalability. The system follows a **three-layer architecture**:

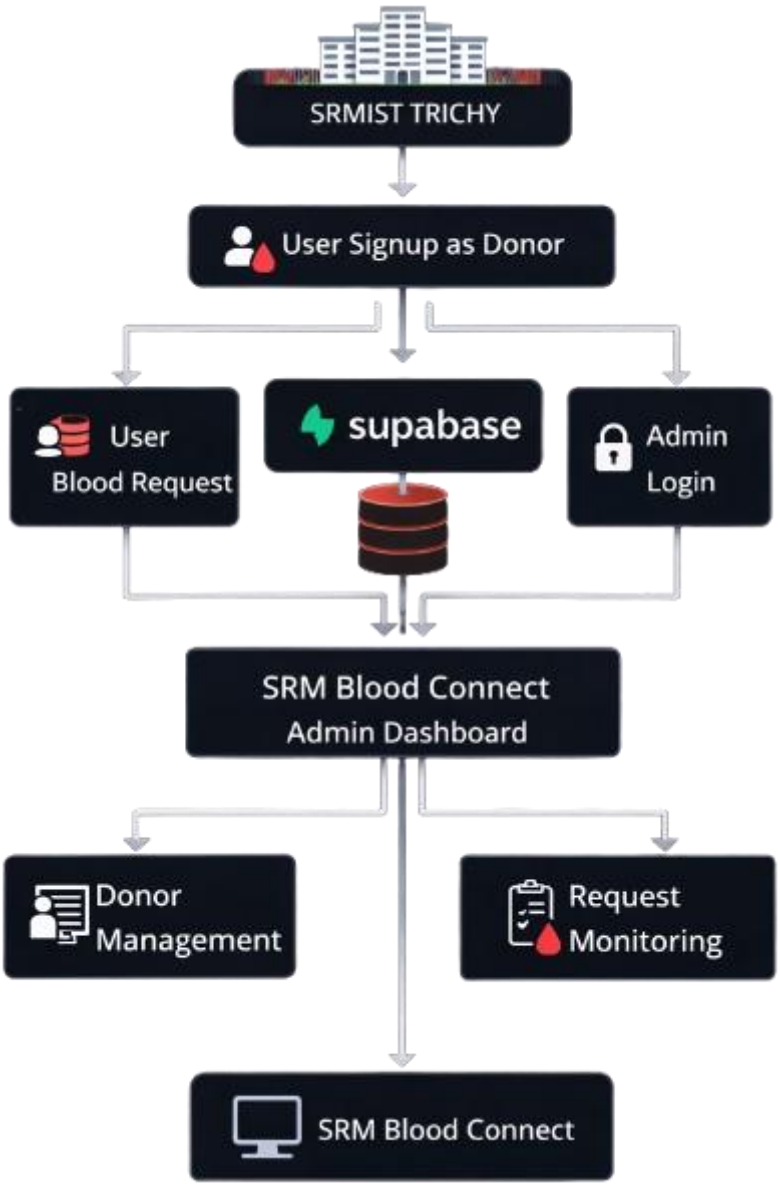

Fig 4.1.1 Architecture Flow

1. Presentation Layer (Frontend)

- Built using **HTML, CSS, JavaScript**

- Contains all user-facing pages such as:
    - Home Page
    - Donor Registration
    - Blood Request Form
    - Login Page
    - Admin Dashboard
- Includes animations, transitions, and responsive layouts
- Validates user inputs before sending data to the backend

2. Application Layer (Logic Layer)

- Handles communication between frontend and backend
- Performs:
    - Form handling
    - Input validation
    - Data formatting
    - API requests
    - Dynamic content rendering
- JavaScript interacts with the Supabase API to send and receive information

3. Data Layer (Database)

- Managed using **Supabase (PostgreSQL)**
- Stores:
    - Donor details
    - Blood group information
    - Contact number
    - Request data

    - Admin login credentials
- Real-time database allows instant updates on donor and request tables
- Supports CRUD operations (Create, Read, Update, Delete)

Architecture Diagram (Description)

- **User** interacts with **Browser UI**
- Browser sends requests to **JavaScript Functions**
- JS communicates with **Supabase API**
- Supabase interacts with **Database**
- Data is returned the same path back to the user

The architecture ensures:

- High performance
- Secure access
- Quick data retrieval
- Simple maintenance

4.2 User Interface Design

The User Interface (UI) of SRM Blood Connect is built to be clean, modern, responsive, and easy to navigate. The design follows a dark theme with glowing UI accents, gradients, and bento-style layouts (from your HTML file).

Key UI Screens

1. Home Page

- Attractive hero section
- Stats, buttons, and navigation menu
- Sidebar and custom cursor features

2. Donor Registration Page

- Form for entering:
    - Name
    - Blood group
    - Phone number
    - Availability
- Simple and clean form design

3. Blood Request Page

- Form to submit request with:
    - Patient name
    - Blood group
    - Reason
    - Contact details

4. Login Page

- Secure admin login
- Simple username/password fields
- Supabase authentication

5. Admin Dashboard

- Sidebar navigation
- Donor list
- Request list
- Real-time updates
- Delete/edit buttons

UI Design Features

- Responsive grid layout

- Smooth transitions and animations
- Custom neon-style buttons
- Glassmorphism elements
- Scroll effects and shadows
- Bento-style cards (from your HTML file)

The UI provides a seamless experience for students, admins, and donors.

4.3 Workflow

The workflow illustrates how users interact with the system and how information flows between components.

A. User Workflow

1. User opens SRM Blood Connect website
2. Chooses to:
    - Register as donor
    - Request blood
3. Fills the form
4. Data is validated
5. Information is sent to database via Supabase
6. Admin receives and reviews data
7. Admin contacts the needed donor/requester
8. Blood donation process is completed

B. Admin Workflow

1. Admin logs into Dashboard
2. Dashboard displays:
    - Registered donors
    - Blood requests
3. Admin verifies information
4. Admin can:
    - Approve/Reject
    - Delete entries
    - Update details
5. Admin contacts appropriate donor/requester

C. Internal Workflow Between Components

1. **Frontend Form → JavaScript**
    - User enters details
    - JS collects and validates data
2. **JavaScript → Supabase API**
    - Data is sent to backend
    - Authentication is checked for admin pages
3. **Supabase → Database**
    - Data is stored in donor or request table
    - Real-time sync triggers updates
4. **Database → Admin Dashboard**
    - Latest entries appear automatically
    - Admin actions update the database

5. PROJECT DESCRIPTION

5.1 Objective

The main objectives of the SRM Blood Connect project are:

1. **To create a centralized digital platform** for managing blood donors and requests within the SRM campus.
2. **To reduce delays** in finding suitable blood donors during emergencies.
3. **To provide a user-friendly system** for registering as a donor or requesting blood.
4. **To digitize and automate the communication process** between donors, recipients, and the admin.
5. **To maintain accurate and verified donor records** using secure backend services.
6. **To enable administrators to manage data easily** through an interactive dashboard.
7. **To promote campus-wide participation** in blood donation and enhance social responsibility.

5.2 Module Dataset Description

1. Module Name: Home Page

- **Input:**
  No direct input (navigation-based)
- **Description:**
  The home page serves as the main interface of the system. It provides an overview of the application and allows users to navigate to different modules such as registration, blood search, emergency alerts, and more.
- **Content:**
  - Navigation bar (Home, Register, Find Blood, etc.)
  - Welcome message
  - Quick access buttons
  - Information banners

2. Module Name: Register Page

- **Input:**
  Name, Age, Gender, Blood Group, Contact Number, Email ID, Address
- **Description:**
  This module allows users to register as blood donors by entering their personal and contact details.
- **Content:**
  - Input fields for user details
  - Dropdowns for Gender and Blood Group
  - Submit button
  - Success message

3. Module Name: Find Blood

- **Input:**
  Blood Group, Location
- **Description:**
  This module helps users find available blood donors based on required blood group and location.
- **Content:**
  - Search filters (Blood Group, Location)
  - Search button
  - List of matching donors with contact details

4. Module Name: Emergency Alert

- **Input:**
  Patient Name, Blood Group, Units Required, Hospital Name, Location, Contact Number

- **Description:**
  This module is used to send urgent blood requests to available donors in case of emergencies.
- **Content:**
  - Input fields for emergency details
  - Submit/Alert button
  - Notification confirmation

5. Module Name: Leaderboard

- **Input:**
  No direct input
- **Description:**
  This module displays top donors based on the number of donations or participation, encouraging users to contribute more.
- **Content:**
  - List of top donors
  - Ranking system
  - Points or donation count display

6. Module Name: Global Search

- **Input:**
  Keywords (Name, Blood Group, Location)
- **Description:**
  This module allows users to search across the entire system for donors, requests, or other relevant information.
- **Content:**
  - Search bar
  - Filter options
  - Display results from multiple modules

7. Module Name: Stats Page

- **Input:**
  No direct input
- **Description:**
  This module provides statistical insights about blood donations, donor availability, and system usage.
- **Content:**
  - Charts and graphs
  - Total donors count
  - Blood group distribution
  - Request statistics

8. Module Name: Notification Page

- **Input:**
  No direct input
- **Description:**
  This module displays alerts and notifications related to emergency requests, updates, and system messages.
- **Content:**
  - List of notifications
  - Alert messages
  - Time/date stamps

9. Module Name: Admin Dashboard

- **Input:**
  Admin login credentials, donor/request data for management
- **Description:**
  This module allows the administrator to manage the entire system including donors, requests, and overall monitoring.

- **Content:**
    - Dashboard overview
    - Donor management (view/edit/delete)
    - Emergency request management
    - Reports and analytics
    - Logout option

Fig 5.2.1 Module Dataset Description

5.3 Module Description

The system is divided into different modules to ensure clarity, efficiency, and maintainability.

1. User Module

- Allows users to register as blood donors.
- Users can submit their details such as name, blood group, phone number, and availability.
- Provides a simple and responsive form interface.

2. Blood Request Module

- Enables users to request blood during emergencies.
- Requires essential details such as patient name, blood group, and contact number.
- Submitted requests are stored in the database for admin review.

3. Admin Authentication Module

- Secure login page for administrators.
- Powered by Supabase authentication.
- Only verified admins can access the dashboard.

4. Admin Dashboard Module

- Displays donor list and request list in separate sections.
- Admin can approve, delete, or update records.
- Real-time database syncing ensures immediate updates.

5. Database Module

- Stores all donated, requested, and user details.
- Uses Supabase PostgreSQL tables.
- Supports CRUD operations (Create, Read, Update, Delete).

6. Notification / Response Module ***(optional for future enhancements)***

- Admin can contact the donor/recipient using stored data.
- Future enhancement: SMS/email alerts for urgent requests.

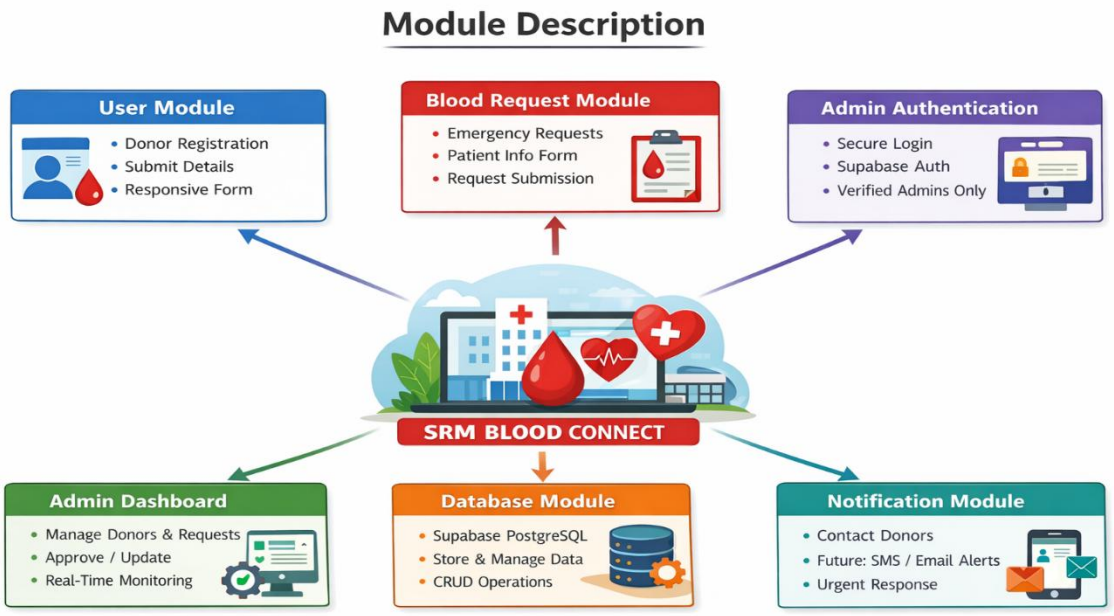

Fig 5.3.1 Module Description

5.4 Implementation

The implementation of SRM Blood Connect involves the use of modern web technologies combined with a cloud-based database service.

Step-by-Step Implementation Approach

1. Frontend Development

- Built using **HTML5, CSS3, JavaScript**.
- UI includes:
    - Home page
    - Registration form
    - Request form
    - Login page
    - Admin dashboard layout

- CSS animations, gradients, and custom cursor enhance the UI design.

2. Supabase Setup

- Created a new project on Supabase.
- Created tables:
    - **donors**
    - **requests**
    - **admin_users**
- Enabled authentication for admin login.
- Configured API keys for JS integration.

3. Database Integration

- JavaScript communicates with Supabase REST API.
- Functions implemented:
    - Adding donor details
    - Submitting blood requests
    - Fetching donor data
    - Deleting/updating entries
    - Admin login verification

4. Admin Panel Implementation

- Sidebar navigation built using HTML/CSS.
- JavaScript fetches and displays donor and request data.
- Admin can delete entries using row-action buttons.
- Real-time updates enabled using Supabase's live query engine.

5. Testing and Debugging

- Verified form validations.

- Checked database operations.
- Ensured UI responsiveness on different screens.
- Fixed layout bugs and alignment issues.

6. Deployment

- Can be hosted on platforms like Netlify, Vercel, or GitHub Pages.
- Supabase remains connected to the deployed frontend.

5.5 Maintenance

Maintenance ensures that the system remains functional, up-to-date, and error-free.

Types of Maintenance

1. Corrective Maintenance

- Fixing bugs or UI issues reported by users.
- Debugging database errors or invalid entries.

2. Adaptive Maintenance

- Updating the system as new requirements emerge.
- Adding new features like notification systems or mobile app support.

3. Perfective Maintenance

- Improving the interface design.
- Enhancing system speed and responsiveness.
- Cleaning old or inactive donor records.

4. Preventive Maintenance

- Regular database backups.
- Checking for security vulnerabilities.
- Monitoring Supabase usage limits.

Overall Maintenance Tasks

- Update donor list periodically.
- Remove duplicate or outdated records.
- Maintain secure admin credentials.
- Ensure forms and UI elements work on all browsers.
- Optimize images, scripts, and CSS for performance.

6. SYSTEM TESTING

6.1 Testing Definition and Objective

Testing Definition

System Testing is the process of evaluating the fully integrated software system to ensure that it meets the specified requirements. It checks the proper interaction of all modules such as donor registration, blood request submission, admin login, and dashboard operations.

Objectives of System Testing

The primary objectives of testing in the SRM Blood Connect project are:

1. **To identify and fix errors** in the functionality, UI, and database operations.
2. **To ensure smooth interaction** between frontend, backend (Supabase), and database.
3. **To verify system behavior** under different conditions and user inputs.
4. **To ensure that all modules work together** as planned without conflicts.
5. **To confirm that the system meets user needs** and project requirements.
6. **To validate system performance**, responsiveness, and reliability.

6.2 Types of Testing

Different types of testing were carried out to ensure complete system validation:

1. Unit Testing

- Tested individual functions such as:
    - Form validation
    - API calls
    - Login authentication
- Ensures each component works independently.

2. Integration Testing

- Checked communication between modules:
    - Donor form → JS functions → Supabase table

    - Admin login → authentication → dashboard

3. Functional Testing

- Verified if features work according to the requirement.
- Examples:
    - Donor registration
    - Blood request submission
    - Real-time updates

4. System Testing

- Entire system tested as a whole.
- Checked navigation, UI responsiveness, page performance.

5. Database Testing

- Checked insertion, deletion, updating, and retrieval of data.
- Verified Supabase real-time syncing.

6. User Interface (UI) Testing

- Ensured all pages display correctly on:
    - Mobile
    - Tablet
    - Laptop/PC
- Checked fonts, colors, buttons, and responsiveness.

7. Performance Testing

- Tested loading speed of forms and dashboard.
- Verified smooth performance even with multiple records.

8. Security Testing

- Checked admin login functionality.

- Verified restricted access to admin features.

6.3 Test Cases

Below is a well-formatted test case table suitable for your report:

Test Case ID	Description	Test Input	Expected Output	Actual Output	Status
TC01	Donor Registration with valid data	Name, blood group, phone	Donor added to database	Successfully added	Pass
TC02	Donor Registration with missing fields	Empty fields	Validation error	Error displayed	Pass
TC03	Blood Request with valid data	Patient name, group, phone	Request stored in DB	Successfully stored	Pass
TC04	Invalid phone number	Wrong format	Show validation message	Validation triggered	Pass
TC05	Admin Login with correct credentials	Valid email & password	Redirect to dashboard	Logged in successfully	Pass
TC06	Admin Login with wrong password	Incorrect password	Show error message	Error displayed	Pass
TC07	Display donor list	Database entries	Show donor data	Donor data displayed	Pass
TC08	Delete donor entry	Delete button	Entry removed	Successfully deleted	Pass
TC09	UI responsiveness	Mobile/Tablet view	Proper layout	Responsive display	Pass
TC10	Real-time sync	Add donor/request	Immediately visible	Synced in dashboard	Pass

7. CONCLUSION

The SRM Blood Connect system successfully provides a reliable digital platform that simplifies the process of blood donor registration and emergency blood requests within the SRM campus. By integrating modern web technologies and a cloud-based backend, the project achieves its goal of connecting donors and recipients quickly and efficiently. The system reduces manual effort, minimizes response time during emergencies, and ensures accurate and accessible information for both users and administrators.

7.1 Summary

The SRM Blood Connect project was developed to address the challenges faced in finding suitable blood donors during urgent situations. The key highlights of the system are:

- A **centralized database** to maintain donor details and blood requests.
- A **user-friendly interface** created using HTML, CSS, and JavaScript.
- **Supabase integration** for authentication, real-time updates, and secure data storage.
- A dedicated **Admin Dashboard** to manage donors, verify requests, and maintain records.
- A responsive layout ensuring **compatibility across devices**.
- Thorough **testing** to verify functionality, performance, and reliability.

Overall, the system meets its intended objectives by offering an efficient, accurate, and organized solution to campus-level blood management.

7.2 Future Enhancements

To further improve the system and increase usability, several enhancements can be added in future versions:

1. **Mobile Application**
    - Android/iOS app for easier access and real-time updates.
2. **Automated Notifications**

- SMS / Email alerts for new requests or urgent blood needs.

3. **AI-Based Donor Matching**
    - Intelligent system to suggest best donors based on availability and compatibility.
4. **Geo-Location Services**
    - Map-based tracking of donors or nearby hospitals.
5. **Donation History & Reward System**
    - Track donor contributions and award digital badges to motivate participation.
6. **Multi-Campus/City Integration**
    - Expand the system to all SRM campuses or partner medical facilities.
7. **Enhanced Security**
    - Add two-factor authentication and advanced data encryption.

7.3 Overall Impact

The SRM Blood Connect project creates a **meaningful impact** on the SRM community:

- **Improved Emergency Response:** Donors can be reached faster, reducing critical delays.
- **Promotes Social Responsibility:** Encourages students and staff to contribute to life-saving activities.
- **Accurate Data Management:** Reduces errors by automating donor and request handling.
- **Ease of Administration:** Minimizes manual work for college officials through the admin dashboard.
- **Strengthened Campus Safety:** Ensures immediate access to verified donor information during emergencies.

Overall, SRM Blood Connect transforms the traditional blood search process into an efficient, reliable, and campus-friendly solution that can potentially save many lives.

8. APPENDIX

8.1 Screenshots

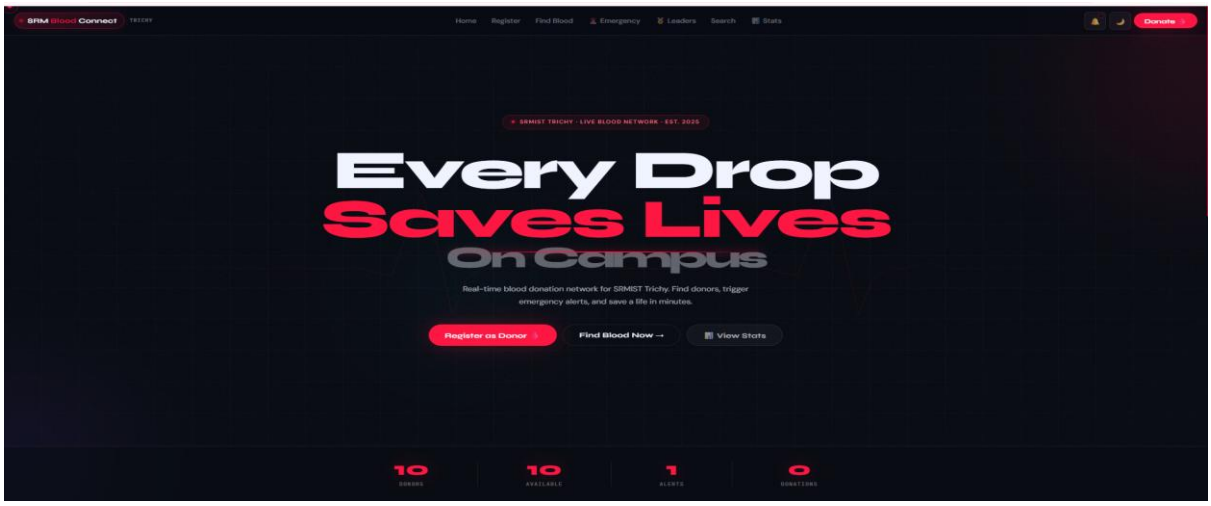

Fig 8.1.1 Home Page

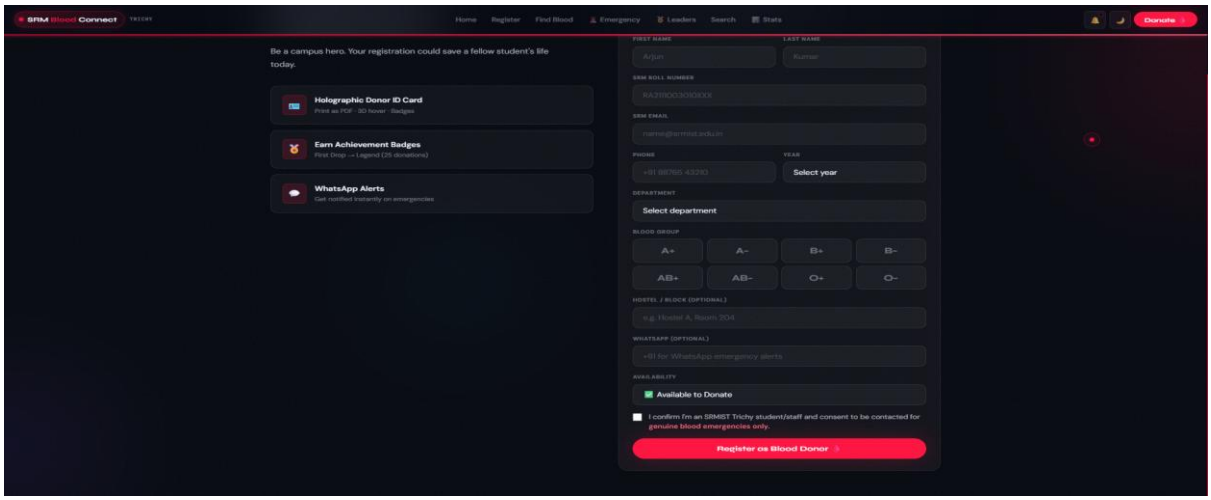

Fig 8.1.2 Register Page

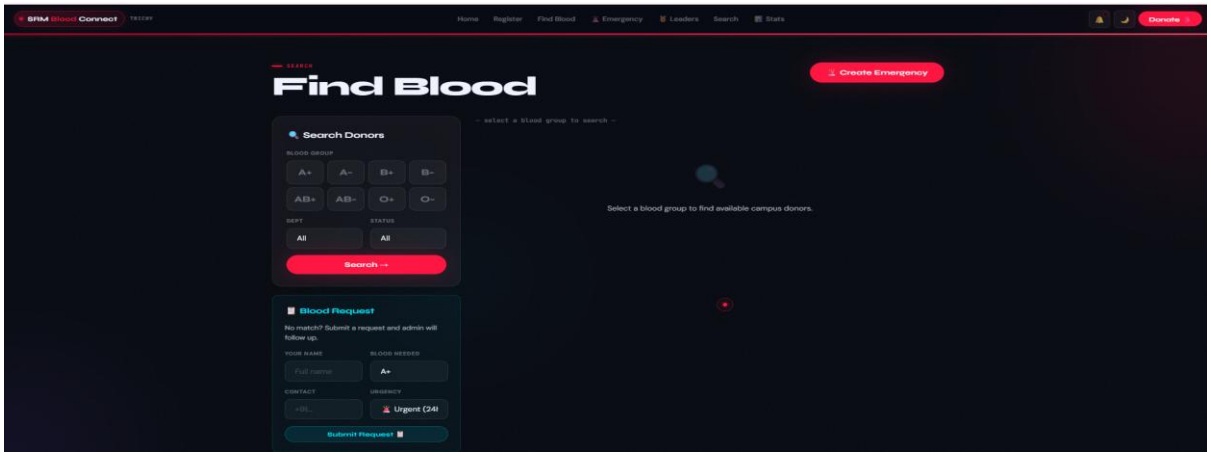

Fig 8.1.3 Find Blood

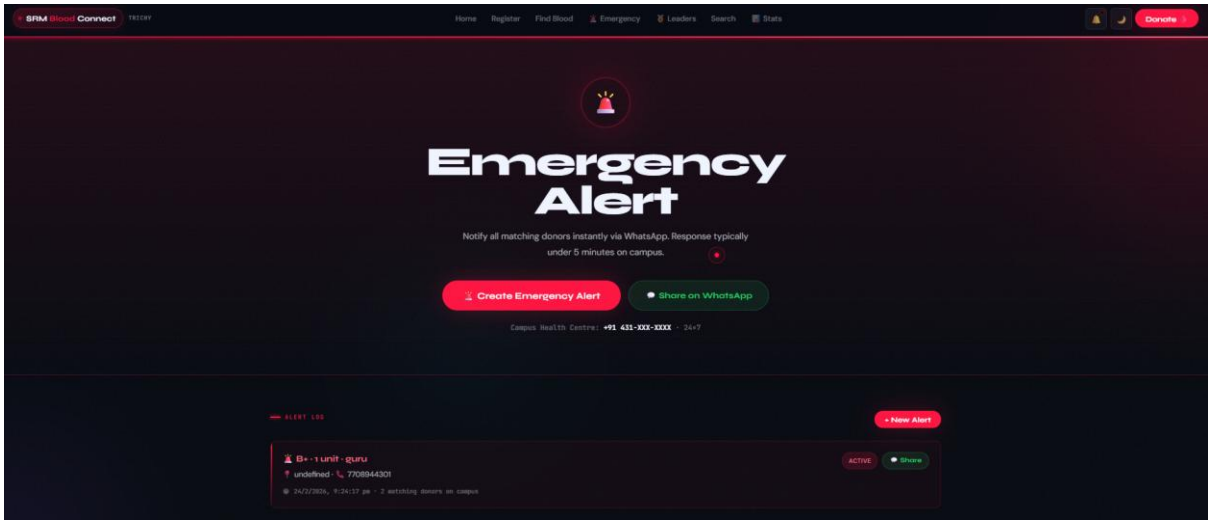

Fig 8.1.4 Emergency Alert

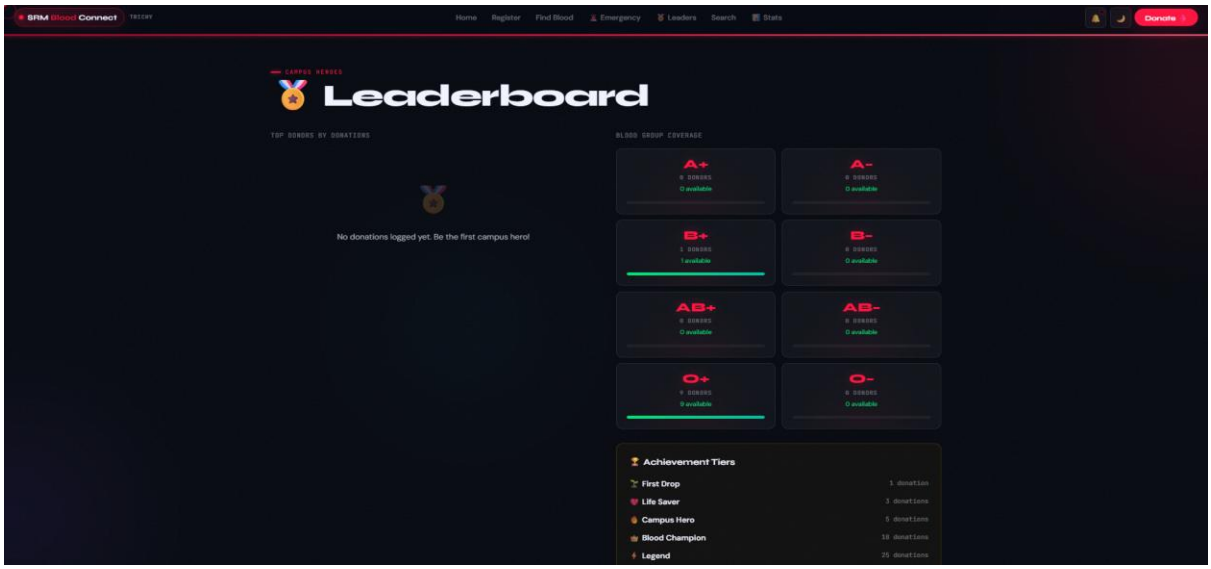

Fig 8.1.5 Leaders Page

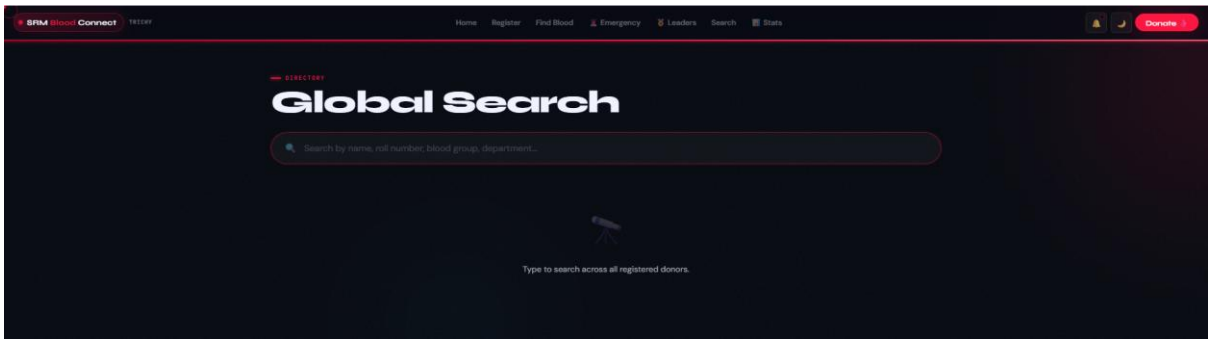

Fig 8.1.6 Global Search

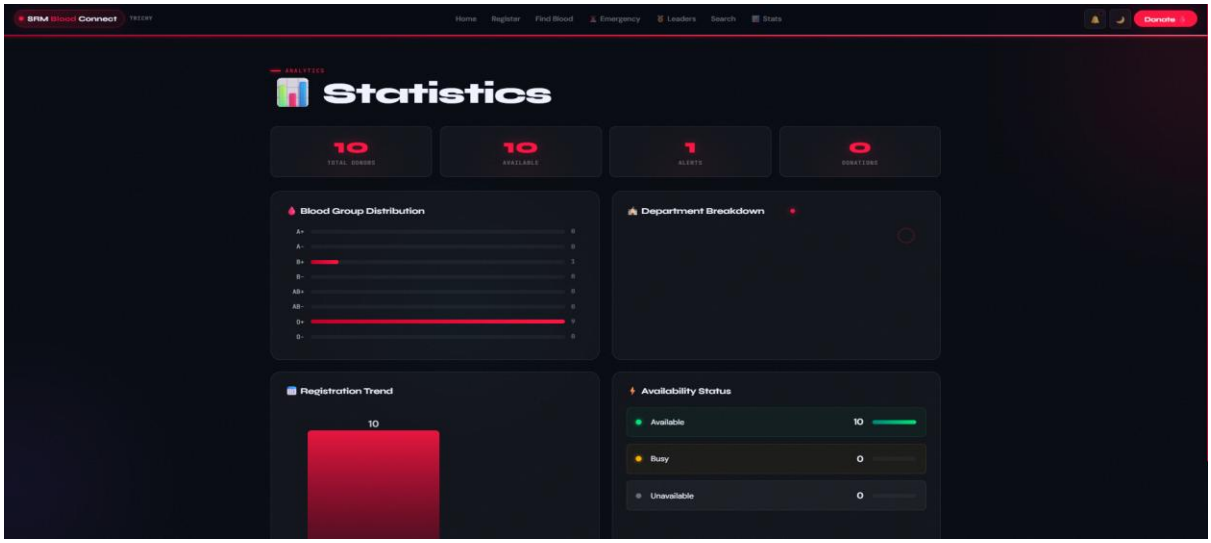

Fig 8.1.7 Stats Page

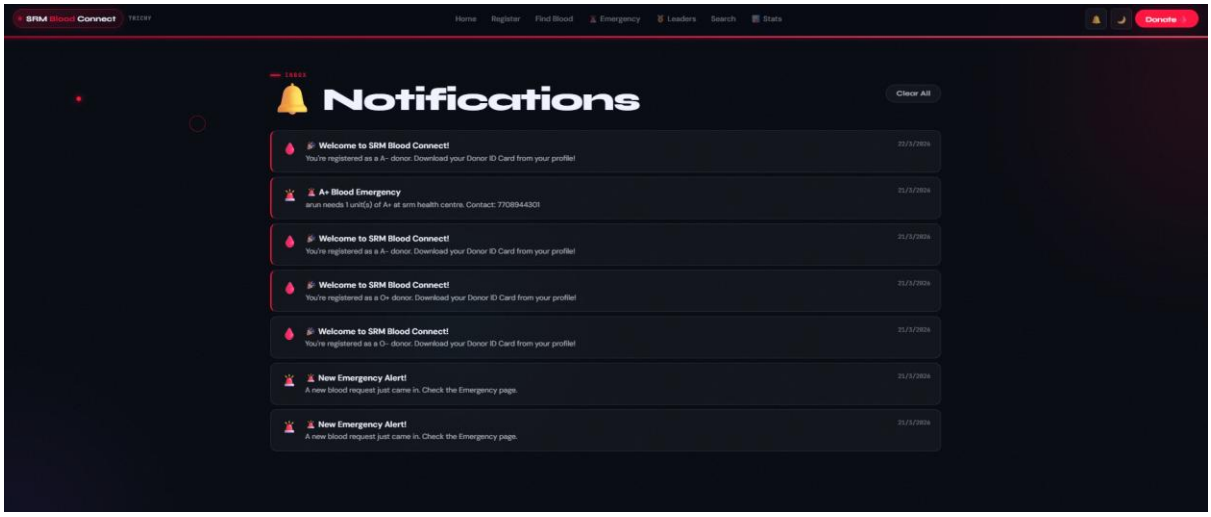

Fig 8.1.8 Notifications Page

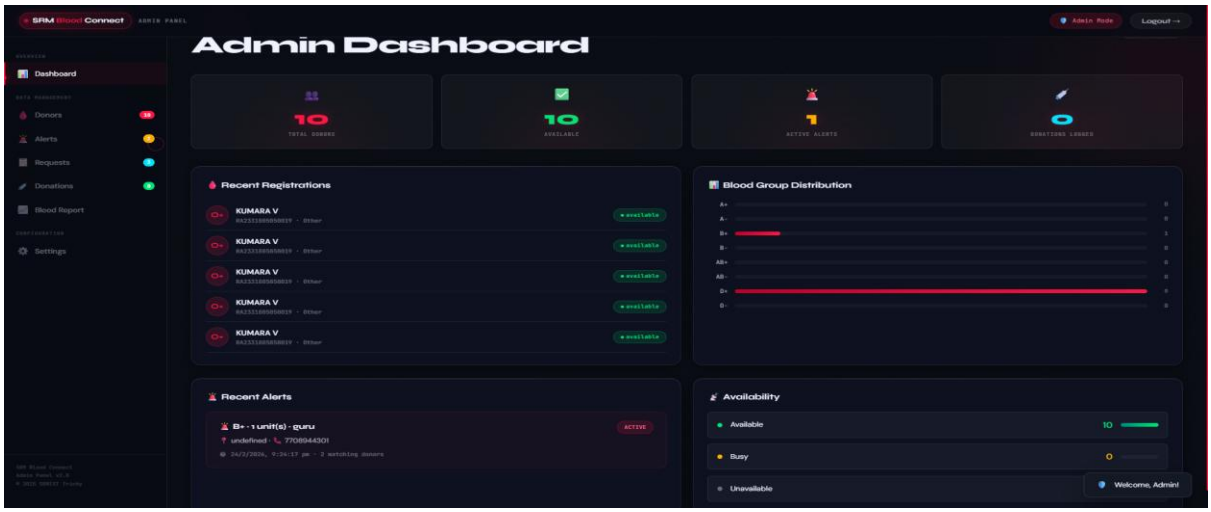

Fig 8.1.9 Admin Dashboard

8.2 CODING

SRM-BLOOD-CONNECT.HTML

```
<!DOCTYPE html>

<html lang="en" data-theme="dark">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1,maximum-scale=1">

<title>SRM Blood Connect · SRMIST Trichy</title>

<link
href="https://fonts.googleapis.com/css2?family=Syne:wght@400;500;600;700;800&family=DM+Sans:ital,opsz,wght@0,9..40,300;0,9..40,400;0,9..40,500;0,9..40,600;1,9..40,400&family=JetBrains+Mono:wght@400;500;700&display=swap" rel="stylesheet">

<script src="https://cdn.jsdelivr.net/npm/@supabase/supabase-
js@2/dist/umd/supabase.min.js"></script>

<script src="https://cdn.jsdelivr.net/npm/html2canvas@1.4.1/dist/html2canvas.min.js"></script>

<style>

/* ═══════════════════════════════════════════════════════════

  DESIGN TOKENS

═══════════════════════════════════════════════════════════ */

:root{

  --red:#FF1744;--red-dim:rgba(255,23,68,.15);--red-glow:rgba(255,23,68,.35);

  --red-dark:#C0002A;--red-soft:#FF6B81;

  --bg:#080B12;--bg2:#0E1220;--bg3:#141926;--bg4:#1A2030;

  --glass:rgba(255,255,255,.04);--glass-border:rgba(255,255,255,.08);--glass-
hover:rgba(255,255,255,.07);

  --t1:#F0F4FF;--t2:rgba(240,244,255,.65);--t3:rgba(240,244,255,.38);--t4:rgba(240,244,255,.18);
```

```
  --cyan:#00E5FF;--cyan-dim:rgba(0,229,255,.12);
  --emerald:#00E676;--emerald-dim:rgba(0,230,118,.12);
  --amber:#FFB300;--amber-dim:rgba(255,179,0,.12);
  --violet:#7C4DFF;--violet-dim:rgba(124,77,255,.12);
  --nav-h:56px;--r:16px;--r-sm:10px;--r-xs:6px;
  --f-display:'Syne',sans-serif;--f-body:'DM Sans',sans-serif;--f-mono:'JetBrains Mono',monospace;
  --shadow-red:0 0 30px rgba(255,23,68,.2),0 0 60px rgba(255,23,68,.08);
  --shadow-card:0 4px 20px rgba(0,0,0,.4),inset 0 1px 0 rgba(255,255,255,.06);
  --shadow-float:0 16px 48px rgba(0,0,0,.5),0 0 0 1px rgba(255,255,255,.06);
  --px:1rem;/* global side padding */
}
[data-theme="light"]{
  --bg:#F5F7FF;--bg2:#EDEFFE;--bg3:#E4E8FF;--bg4:#D9DFFF;
  --glass:rgba(0,0,0,.03);--glass-border:rgba(0,0,0,.08);--glass-hover:rgba(0,0,0,.05);
  --t1:#0A0F2E;--t2:rgba(10,15,46,.65);--t3:rgba(10,15,46,.4);--t4:rgba(10,15,46,.15);
}
/* ═══════════════════════════════════════════════════════════
   RESET & BASE
═══════════════════════════════════════════════════════════ */
*,*::before,*::after{box-sizing:border-box;margin:0;padding:0}
html{scroll-behavior:smooth;-webkit-text-size-adjust:100%}
body{font-family:var(--f-body);background:var(--bg);color:var(--t1);overflow-x:hidden;-webkit-font-smoothing:antialiased;cursor:none;line-height:1.6;font-size:15px}
a,button,[onclick]{cursor:none}
::-webkit-scrollbar{width:2px}::-webkit-scrollbar-track{background:transparent}::-webkit-scrollbar-thumb{background:var(--red);border-radius:1px}
```

```
img,svg{display:block;max-width:100%}

input,textarea,select{font-family:var(--f-body)}


/* ═══════════════════════════════════════════════════════════

   CUSTOM CURSOR (desktop only)

═══════════════════════════════════════════════════════════ */

#cur{position:fixed;inset:0;pointer-events:none;z-index:9999}

#cur-dot{position:absolute;width:7px;height:7px;background:var(--red);border-
radius:50%;transform:translate(-50%,-50%);box-shadow:0 0 10px var(--red)}

#cur-ring{position:absolute;width:26px;height:26px;border:1.5px solid rgba(255,23,68,.4);border-
radius:50%;transform:translate(-50%,-50%);transition:transform .16s cubic-bezier(.2,.8,.2,1),width
.25s,height .25s}

@media(hover:none){#cur{display:none}body,a,button,[onclick]{cursor:auto}}


/* ═══════════════════════════════════════════════════════════

   NOISE + BLOBS

═══════════════════════════════════════════════════════════ */

body::before{content:'';position:fixed;inset:0;z-index:1;opacity:.025;pointer-events:none;

  background-image:url("data:image/svg+xml,%3Csvg viewBox='0 0 256 256'
xmlns='http://www.w3.org/2000/svg'%3E%3Cfilter id='n'%3E%3CfeTurbulence type='fractalNoise'
baseFrequency='.9' numOctaves='4' stitchTiles='stitch'/%3E%3C/filter%3E%3Crect width='100%25'
height='100%25' filter='url(%23n)'/%3E%3C/svg%3E");}

.blobs{position:fixed;inset:0;z-index:0;pointer-events:none;overflow:hidden}

.blob{position:absolute;border-radius:50%;filter:blur(80px);opacity:.14;animation:morph 20s ease-in-
out infinite alternate}

.blob-1{width:min(600px,80vw);height:min(600px,80vw);background:radial-
gradient(circle,#FF1744,transparent 70%);top:-20%;right:-15%;animation-duration:18s}
```

```
.blob-2{width:min(400px,60vw);height:min(400px,60vw);background:radial-gradient(circle,#7C4DFF,transparent 70%);bottom:-10%;left:-10%;animation-duration:22s;animation-delay:-8s}

.blob-3{width:min(300px,50vw);height:min(300px,50vw);background:radial-gradient(circle,#00E5FF,transparent 70%);top:40%;left:30%;animation-duration:25s;animation-delay:-12s;opacity:.06}

@keyframes morph{0%{transform:translate(0,0) scale(1)}50%{transform:translate(3%,5%) scale(1.08)}100%{transform:translate(-2%,-3%) scale(.96)}}

</html>
```

SRM-ADMIN-PANNEL.HTML

```
<!DOCTYPE html>

<html lang="en" data-theme="dark">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1,maximum-scale=1">

<title>Admin Panel · SRM Blood Connect</title>

<link
href="https://fonts.googleapis.com/css2?family=Syne:wght@400;500;600;700;800&family=DM+Sans:ital,opsz,wght@0,9..40,300;0,9..40,400;0,9..40,500;0,9..40,600;1,9..40,400&family=JetBrains+Mono:wght@400;500;700&display=swap" rel="stylesheet">

<script src="https://cdn.jsdelivr.net/npm/@supabase/supabase-js@2/dist/umd/supabase.min.js"></script>

<style>

/* ═══ DESIGN TOKENS ═══ */

:root{

  --red:#FF1744;--red-dim:rgba(255,23,68,.15);--red-glow:rgba(255,23,68,.35);

  --red-dark:#C0002A;--red-soft:#FF6B81;
```

```
  --bg:#080B12;--bg2:#0E1220;--bg3:#141926;--bg4:#1A2030;

  --glass:rgba(255,255,255,.04);--glass-border:rgba(255,255,255,.08);--glass-
hover:rgba(255,255,255,.07);

  --t1:#F0F4FF;--t2:rgba(240,244,255,.65);--t3:rgba(240,244,255,.38);--t4:rgba(240,244,255,.18);

  --cyan:#00E5FF;--cyan-dim:rgba(0,229,255,.12);

  --emerald:#00E676;--emerald-dim:rgba(0,230,118,.12);

  --amber:#FFB300;--amber-dim:rgba(255,179,0,.12);

  --violet:#7C4DFF;

  --nav-h:60px;--sidebar-w:240px;--r:16px;--r-sm:10px;--r-xs:6px;

  --f-display:'Syne',sans-serif;--f-body:'DM Sans',sans-serif;--f-mono:'JetBrains Mono',monospace;

  --shadow-red:0 0 30px rgba(255,23,68,.2),0 0 60px rgba(255,23,68,.08);

  --shadow-card:0 4px 20px rgba(0,0,0,.4),inset 0 1px 0 rgba(255,255,255,.06);

  --shadow-float:0 16px 48px rgba(0,0,0,.5),0 0 0 1px rgba(255,255,255,.06);

}


/* ═══ RESET ═══ */

*,*::before,*::after{box-sizing:border-box;margin:0;padding:0}

html{scroll-behavior:smooth}

body{font-family:var(--f-body);background:var(--bg);color:var(--t1);overflow-x:hidden;-webkit-font-
smoothing:antialiased;line-height:1.6;font-size:15px;cursor:none}

a,button,[onclick]{cursor:none}

::-webkit-scrollbar{width:3px}::-webkit-scrollbar-track{background:transparent}::-webkit-scrollbar-
thumb{background:var(--red);border-radius:2px}

input,select,textarea{font-family:var(--f-body)}

@media(hover:none){body,a,button,[onclick]{cursor:auto}}


/* ═══ NOISE ═══ */
```

```
body::before{content:'';position:fixed;inset:0;z-index:1;opacity:.02;pointer-events:none;
  background-image:url("data:image/svg+xml,%3Csvg viewBox='0 0 256 256'
xmlns='http://www.w3.org/2000/svg'%3E%3Cfilter id='n'%3E%3CfeTurbulence type='fractalNoise'
baseFrequency='.9' numOctaves='4' stitchTiles='stitch'/%3E%3C/filter%3E%3Crect width='100%25'
height='100%25' filter='url(%23n)'/%3E%3C/svg%3E");}
.blobs{position:fixed;inset:0;z-index:0;pointer-events:none;overflow:hidden}
.blob{position:absolute;border-radius:50%;filter:blur(90px);opacity:.1;animation:morph 20s ease-in-
out infinite alternate}
.blob-1{width:500px;height:500px;background:radial-gradient(circle,#FF1744,transparent 70%);top:-
15%;right:-10%;animation-duration:18s}
.blob-2{width:350px;height:350px;background:radial-gradient(circle,#7C4DFF,transparent
70%);bottom:-10%;left:-8%;animation-duration:22s;animation-delay:-8s}
@keyframes morph{0%{transform:translate(0,0) scale(1)}100%{transform:translate(2%,4%)
scale(1.06)}}
@keyframes pulse-
live{0%,100%{opacity:1;transform:scale(1)}50%{opacity:.5;transform:scale(.8)}}

/* ═══ CURSOR ═══ */
#cur{position:fixed;inset:0;pointer-events:none;z-index:9999}
#cur-dot{position:absolute;width:7px;height:7px;background:var(--red);border-
radius:50%;transform:translate(-50%,-50%);box-shadow:0 0 10px var(--red)}
#cur-ring{position:absolute;width:26px;height:26px;border:1.5px solid rgba(255,23,68,.4);border-
radius:50%;transform:translate(-50%,-50%);transition:transform .16s cubic-bezier(.2,.8,.2,1),width
.25s,height .25s}
@media(hover:none){#cur{display:none}}

/* ═══ TOP NAV ═══ */
.topnav{position:fixed;top:0;left:0;right:0;height:var(--nav-h);z-index:400;
  display:flex;align-items:center;justify-content:space-between;padding:0 1.5rem;
```

```
  background:rgba(8,11,18,.85);backdrop-filter:blur(40px) saturate(2);
  border-bottom:1px solid var(--glass-border)}
.nav-brand{display:flex;align-items:center;gap:.65rem}
.brand-pill{display:flex;align-items:center;gap:.4rem;padding:.28rem .75rem .28rem .45rem;
  background:linear-gradient(135deg,rgba(255,23,68,.2),rgba(255,23,68,.08));
  border:1px solid rgba(255,23,68,.3);border-radius:50px;box-shadow:0 0 16px rgba(255,23,68,.12)}
.brand-pulse{width:7px;height:7px;border-radius:50%;background:var(--red);box-shadow:0 0 6px
var(--red);animation:pulse-live 1.8s ease-in-out infinite;flex-shrink:0}
.brand-text{font-family:var(--f-display);font-size:.82rem;font-weight:700;color:var(--t1)}
.brand-text em{color:var(--red);font-style:normal}
.brand-sub{font-size:.6rem;font-weight:700;letter-spacing:.14em;text-transform:uppercase;color:var(-
-t3);font-family:var(--f-mono)}
.nav-right{display:flex;align-items:center;gap:.8rem}
.nav-badge{display:flex;align-items:center;gap:.4rem;padding:.3rem .8rem;border-
radius:50px;background:var(--red-dim);border:1px solid rgba(255,23,68,.2);font-size:.68rem;font-
family:var(--f-mono);color:var(--red-soft)}
.btn-logout{padding:.42rem 1.1rem;border-radius:50px;font-family:var(--f-display);font-
size:.75rem;font-weight:700;border:1px solid var(--glass-border);background:var(--glass);color:var(--
t2);cursor:none;transition:all .22s}
.btn-logout:hover{border-color:rgba(255,23,68,.3);color:var(--red);background:var(--red-dim)}
#realtime-dot{width:7px;height:7px;border-radius:50%;background:var(--emerald);box-shadow:0 0
6px var(--emerald);animation:pulse-live 2s infinite;display:none}
#realtime-dot.on{display:inline-block}
```

CODING SCREENSHOTS

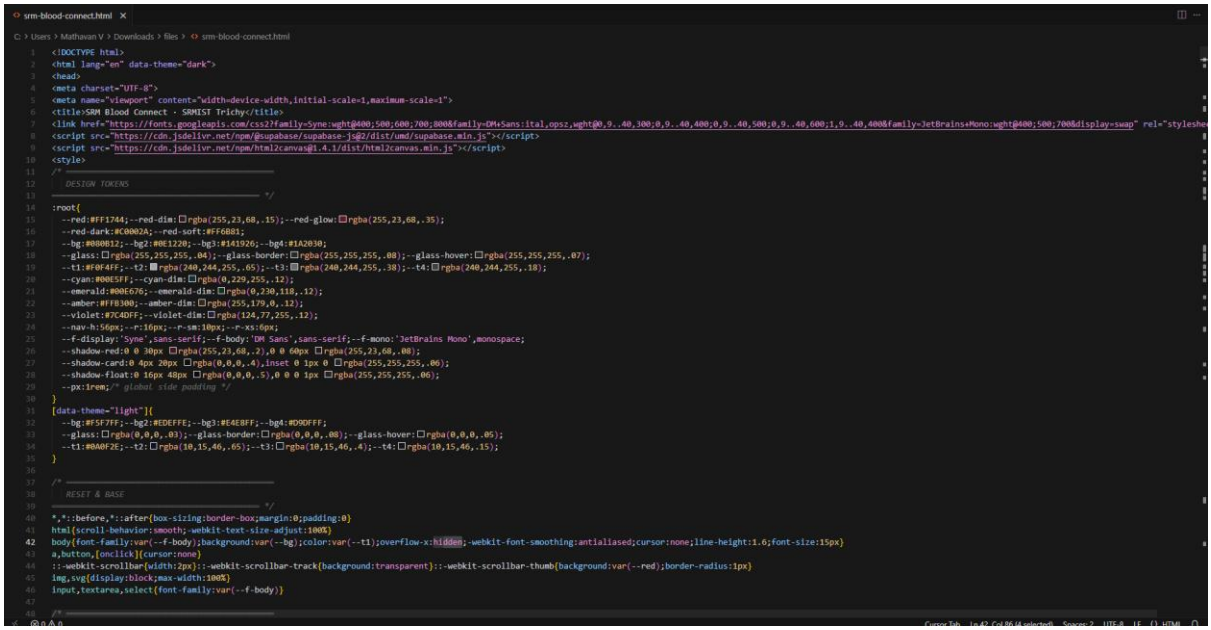

Fig 8.2.1

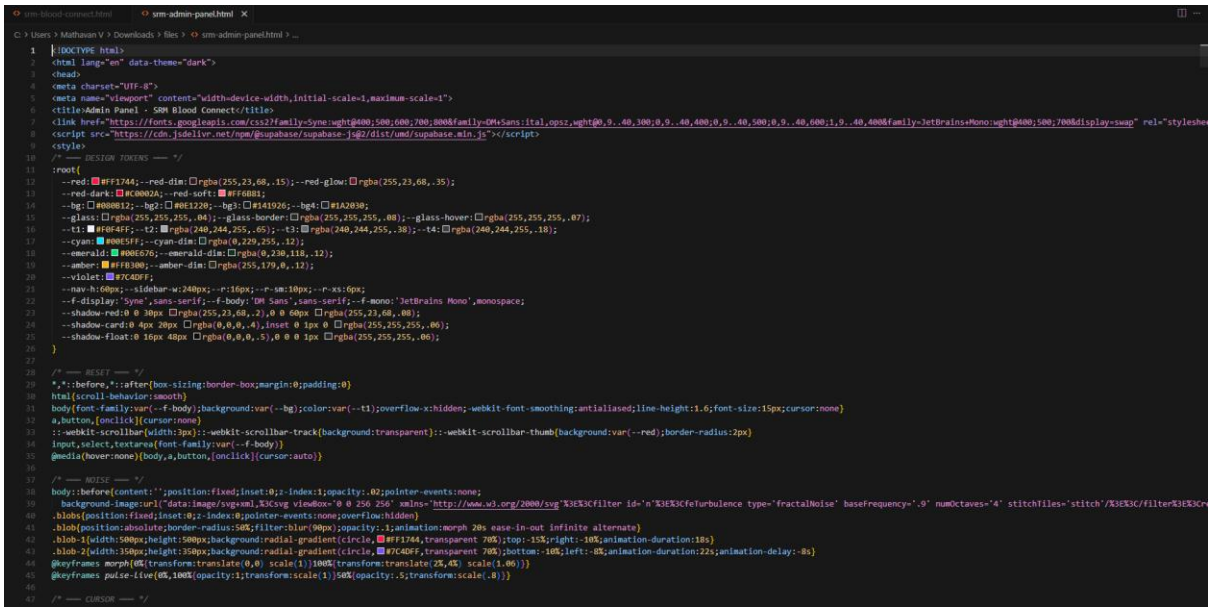

Fig 8.2.2

9. BIBLIOGRAPHY AND REFERENCES

A. Books

1. Rajkumar Buyya, Christian Vecchiola, Thamarai Selvi – *Mastering Cloud Computing*, McGraw Hill.
2. E. Balagurusamy – *Programming with JavaScript*, Tata McGraw-Hill.
3. Achyut S. Godbole & Atul Kahate – *Web Technologies*, McGraw Hill.
4. Thomas Connolly & Carolyn Begg – *Database Systems: A Practical Approach to Design, Implementation and Management*.
5. Roger S. Pressman – *Software Engineering: A Practitioner's Approach*.

B. Research Papers & Journals

1. S. Singh, "Modern Approaches to Blood Donor Management Systems", *International Journal of Computer Applications*.
2. P. Kandukuri, "Web-Based Healthcare Systems for Emergency Response", *IJERT*.
3. L. Ramya & S. Kumar, "Cloud-Based Applications for College-Level Data Management", *JIRSET*.

C. Websites

1. **Supabase Documentation** – https://supabase.com/docs
2. **W3Schools – HTML/CSS/JavaScript** – https://www.w3schools.com
3. **MDN Web Docs – JavaScript & Web APIs** – https://developer.mozilla.org
4. **Stack Overflow** – https://stackoverflow.com
5. **CSS Tricks** – https://css-tricks.com

D. Tools & Platforms Used

1. Visual Studio Code – Official Documentation
2. Supabase (Open-Source Backend) – API Reference
3. Google Chrome Developer Tools
4. GitHub – Repository Management

E. Project-Related Resources

1. SRM Institute of Science and Technology – Internal Project Guidelines
2. Sample datasets for blood donor management systems